

FailMem Supplementary Materials

FailMem: Supplementary Materials & Verification Audit Details

Title: Auditable Failure Memory for ROS 2 Navigation: Supplementary Materials

Document Status: Accompanying Material for Review & Reproduction

Associated Commit: e3cce9d

S1. Precondition Integrity & Doorway Clearance Details

The passage state evaluation in `src/doorway_evaluator.py` enforces strict sensor-TF alignment rules prior to emitting a clearance classification:

S1.1 Precondition Rules

1. **Finite Transforms:** TF translation vector $\mathbf{t} = [x, y, z]^T$ and yaw angle ψ must be finite real numbers ($-\infty < x, y, z, \psi < \infty$).
2. **Temporal Alignment:** Scan and TF timestamps must satisfy $|t_{\text{tf}} - t_{\text{scan}}| \leq 0.80 \text{ s}$.
3. **Data Freshness:** Current simulation clock t_{sim} must satisfy $t_{\text{sim}} - t_{\text{tf}} \leq 1.0 \text{ s}$ and $t_{\text{sim}} - t_{\text{scan}} \leq 1.5 \text{ s}$.
4. **Ray Projections:**
 - Number of finite laser scan ranges $N_{\text{valid}} > 0$.
 - Number of projected rays intersecting the doorway bounding volume $N_{\text{intersect}} > 0$.
5. **Costmap Availability:** When costmap subgrids are provided, total cells in the Region of Interest (ROI) $N_{\text{total_cells}} > 0$.

If any precondition fails, the evaluator emits $\text{State}(B) = \text{UNKNOWN}$ and does not alter memory state.

S1.2 Bounding Volume & Threshold Parameters

- Doorway center: $x = 0.0 \text{ m}, y = 1.20 \text{ m}$.
- ROI bounding box: $x \in [-0.15, 0.15] \text{ m}, y \in [0.95, 1.45] \text{ m}$.
- Minimum pass-through rays for `FREE`: $N_{\text{pass_through}} \geq 8$.
- Maximum hits inside for `FREE`: $N_{\text{hits_inside}} = 0$.
- Hits threshold for `OCCUPIED`: $N_{\text{hits_inside}} \geq 5$ or $N_{\text{costmap_occupied_cells}} > 0$.

S2. Cryptographic Replay Audit Engine

To avoid reliance on self-reported runtime metrics, the independent replay auditor (`scripts/replay_and_score_p2c.py`) re-scores every physical simulation run offline:

S2.1 Replay Integrity Verification Steps

1. **Manifest Audit:** Verifies every serialized file against `checksums.sha256`.
2. **Trajectory Distance:** Computes traversed distance by summing Euclidean distances between consecutive odometry sample positions $\langle x_k, y_k \rangle_{k=1}^K$:

$$d = \sum_{k=2}^K \sqrt{(x_k - x_{k-1})^2 + (y_k - y_{k-1})^2}$$

3. **Halt Stability Verification:** Verifies that upon goal completion, the robot remains stationary over a minimum nominal window of 2.0 s:

$$\max_{t \in [t_{\text{goal}}, t_{\text{goal}} + \Delta t_{\text{stab}}]} |v_{\text{lin}}(t)| \leq 0.05 \text{ m/s}, \quad \max_{t \in [t_{\text{goal}}, t_{\text{goal}} + \Delta t_{\text{stab}}]} |v_{\text{ang}}(t)| \leq 0.08 \text{ rad/s}$$

4. **Chronological Causality:**

$$t_{\text{rec}} \geq \max(t_{\text{action_end}}, t_{\text{obs_eval}}) - 0.05 \text{ s}$$

$$t_{\text{inv}} \geq \max(t_{\text{failure}}, t_{\text{free_eval}}) - 0.05 \text{ s}$$

S3. Machine-Independent PDF Build Pipeline

The manuscript build pipeline (`paper/scripts/build_paper_pdf.py`): 1. **Dynamic Table Injection:** Substitutes `<!-- TABLE:... -->` markers with freshly generated LaTeX and Markdown tables. 2. **MathJax Vector SVG Pre-Rendering:** Converts LaTeX formulas into inline SVG vector paths via Node.js `mathjax-full@3.2.2` resolved from project root `package.json`. 3. **Table Row Validation:** Verifies generated HTML table structure prior to PDF compilation: - Table 1 (Condition Summary): Exactly 10 data rows. - Table 2 (Pairwise Contrasts): Exactly 10 data rows. - Table 3 (H_1 Feasibility): Exactly 4 data rows. 4. **Zero Raw TeX Leak Assertion:** Renders compiled PDF pages to PNG at 150 dpi (`pdftoppm`) and extracts text (`pdftotext`); raises `RuntimeError` on unrendered TeX control sequences. 5. **Structured Build Record:** Emits `paper/build_report.json` documenting tool versions, pre-build git status, post-build workspace changes, and cryptographic input/output digests.

S4. Detailed Parameter & Source Code Traceability

For exact code symbol and line-number mappings for all mathematical formulations and protocol parameters, refer to [paper/method-code-map.csv](#) (33 parameter mappings). For the complete literature primary verification ledger, refer to [paper/reference-verification.csv](#) (15 primary-source verified references).